

File Paths

Session Objectives

By the end of this session, you'll be able to:

- Identify how your computer sorts information via file paths
- Define relative and absolute file paths
- Apply concepts of file paths to importing, or “reading” data into R

What is a file path?

Definitions

- **Files:**
 - Objects on a computer that store data, information, settings, or commands used with a computer program.
- **Directories:**
 - Also called "folders", are the units that hold files or other directories.

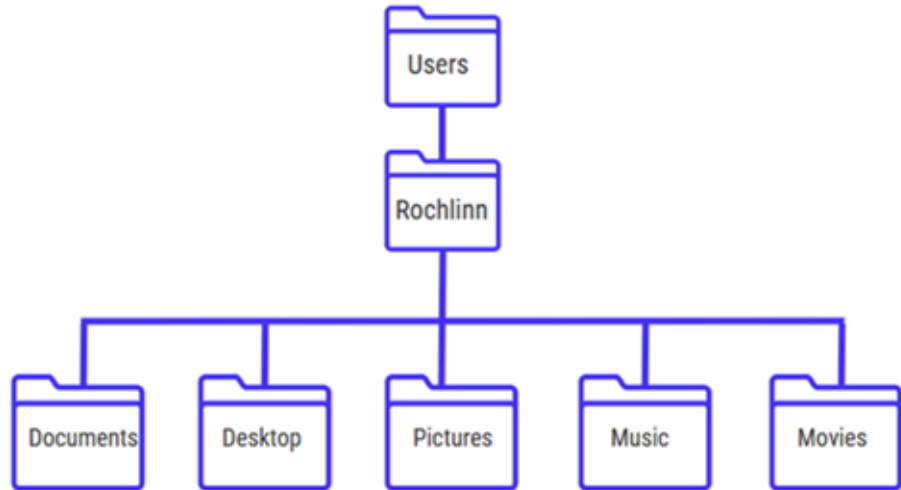
Definitions

- **Paths:**
 - A string of characters that specifies a unique location in a directory hierarchy.
 - A path specifies the location of a directory/file in a machine's file system structure.

Paths

- The picture below illustrates the directory structure of a typical computer.
- Within this structure, the path to the Desktop would be:

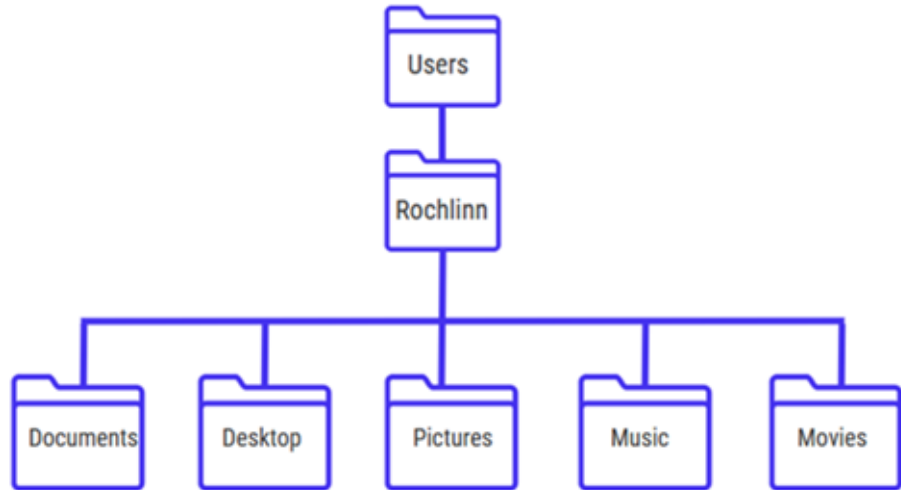
`/Users/Rochlinn/Desktop`



Paths

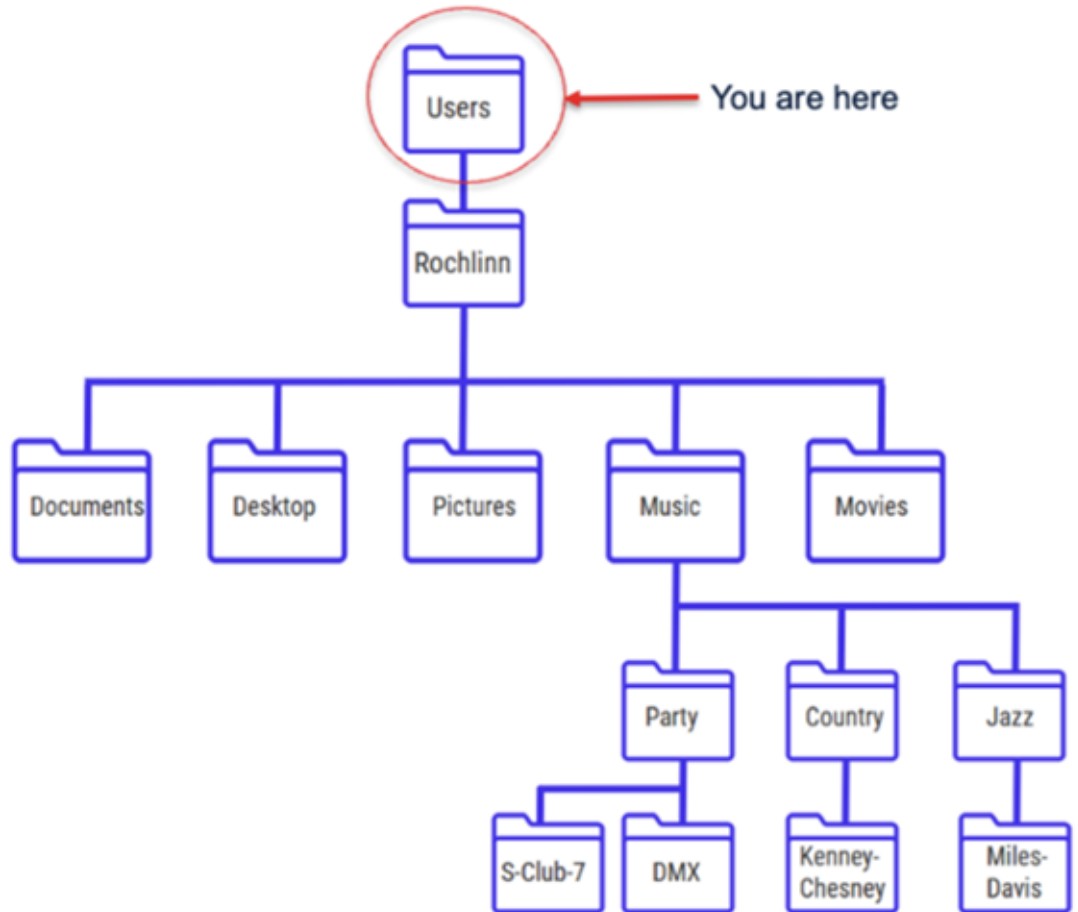
- When dealing with file paths, directories are denoted by a slash /, and each layer" of the directory hierarchy is separated by a slash.

/Users/Rochlinn/Desktop



Exercise

- 1) What is the path to the Music directory?
- 1) What is the path to the S-Club-7 directory?



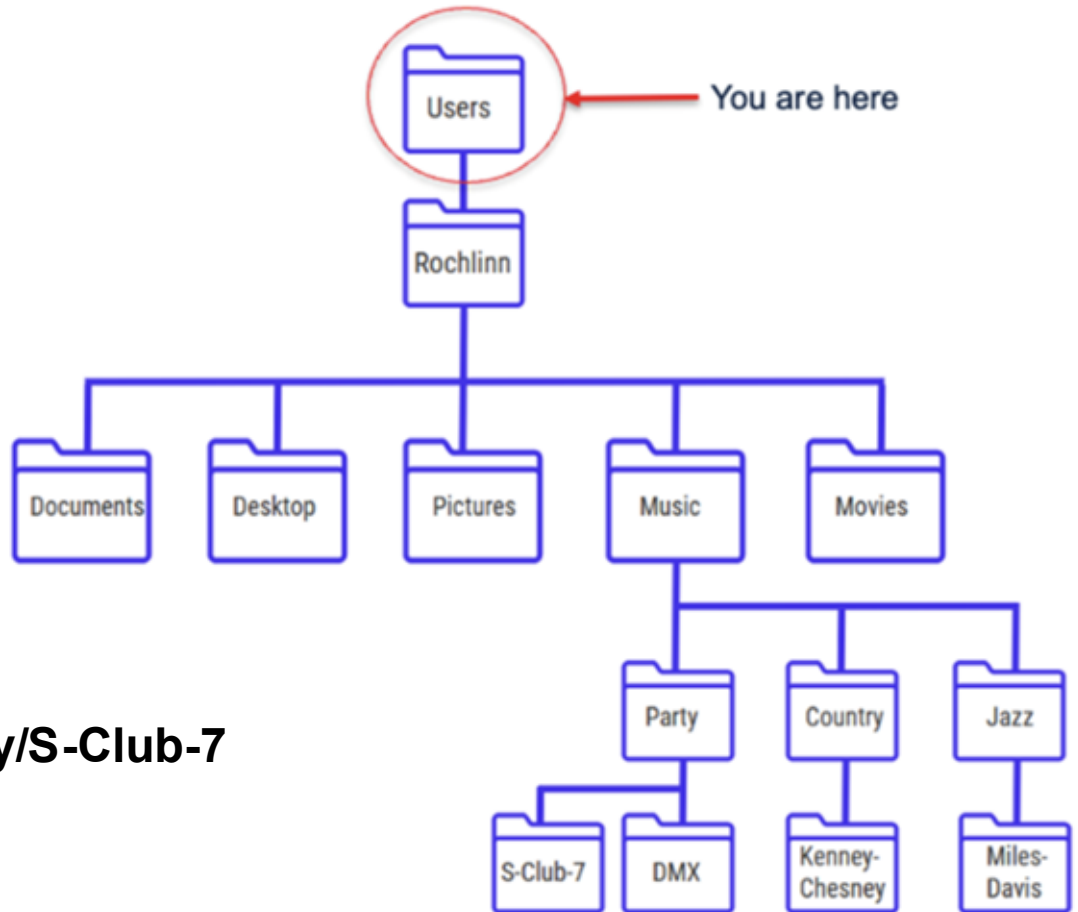
Exercise

- 1) What is the path to the Music directory?

/Users/Rochlinn/Music

- 1) What is the path to the S-Club-7 directory?

/Users/Rochlinn/Music/Party/S-Club-7



Absolute and Relative Paths

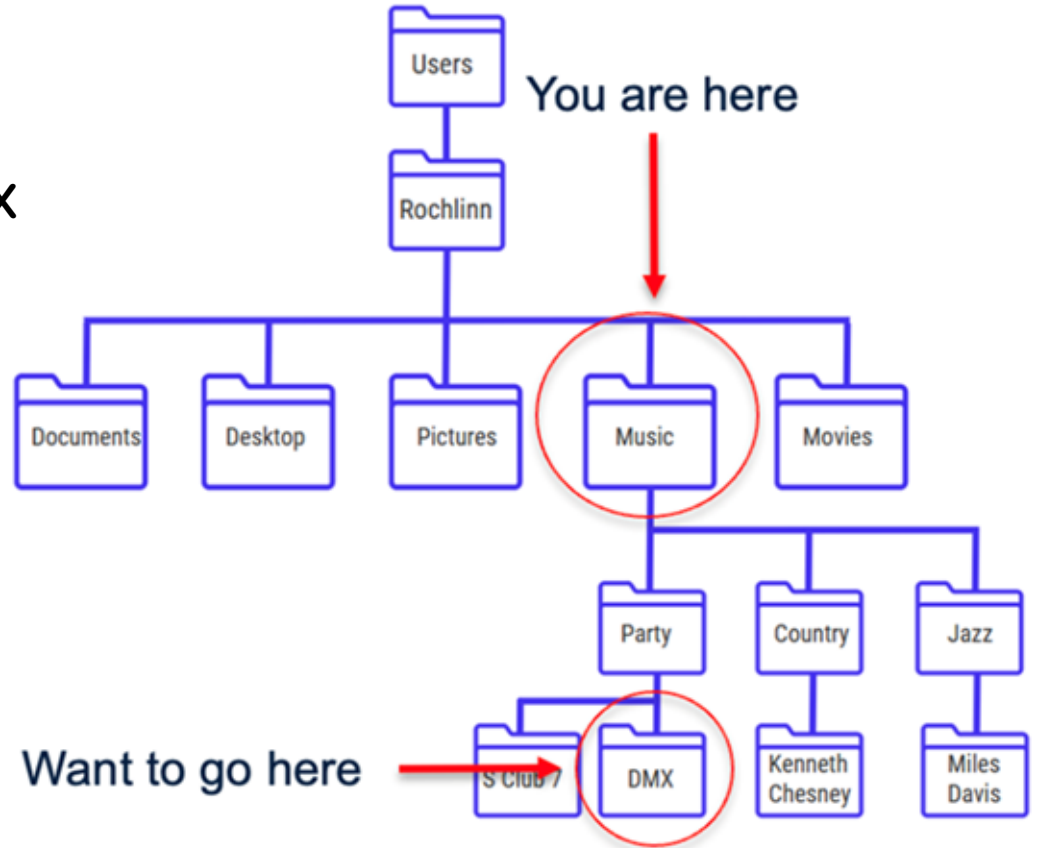
Absolute and Relative Paths

When thinking about navigating through your files and directories, there are two kinds of paths that can be used:

- **Absolute paths:** Includes the entire path from the root directory.
- **Relative paths:** Includes the path relative to where you are in the hierarchy.

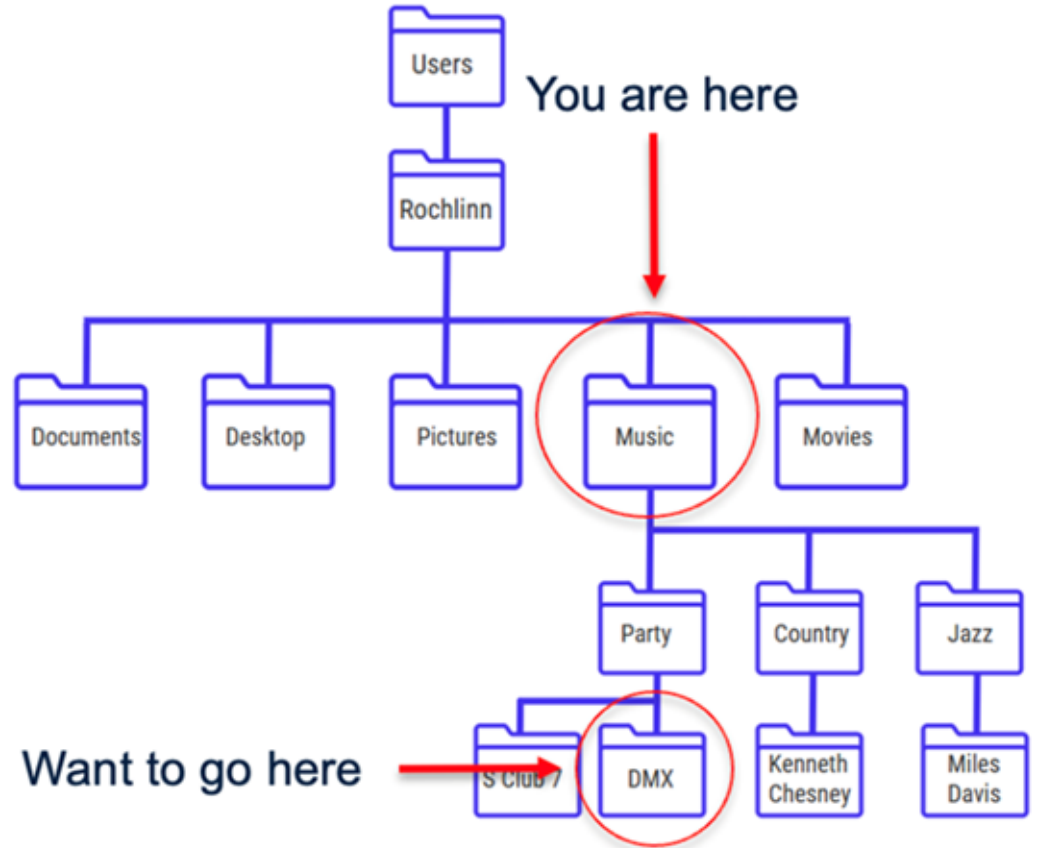
Absolute Paths

`/Users/Rochlinn/Music/Party/DMX`



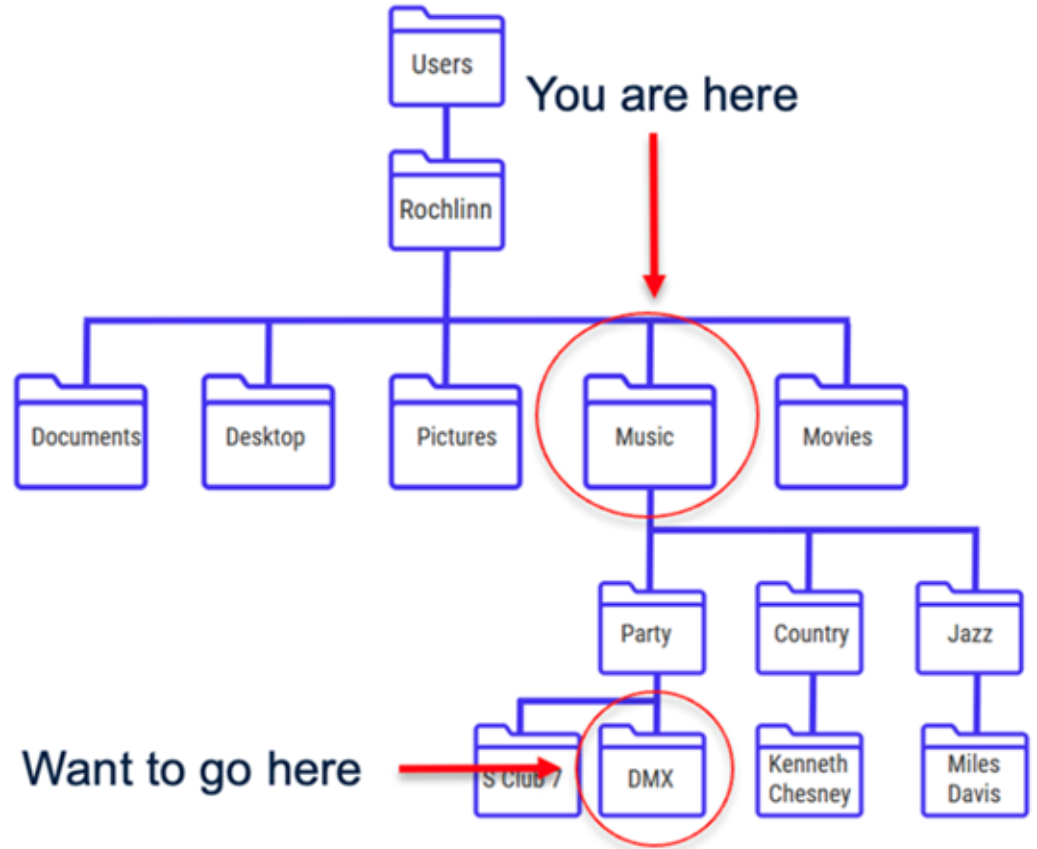
Relative Paths

/Party/DMX



Note

Relative paths only work **going down** the hierarchy, and not upwards.



Why should you care?

- So far we have been playing around with R, but we haven't yet brought data into R.
- When reading data into RStudio, file paths tend to be a key point of struggle/issues for those that are new to coding.
- There isn't a need to be an expert in file paths, but this session is meant to provide the necessary information needed to engage with the next session, in which we'll be importing data into R.

Questions?